

Registration no:

--	--	--	--	--	--	--	--	--	--

Total Number of Pages: 02

B.TECH
PCEL4301

5th Semester Regular / Back Examination 2016-17

POWER ELECTRONICS

BRANCH(S): EE,EEE

Time: 3 Hours

Max Marks: 70

Q.CODE: Y128

**Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.**

Q1 Answer the following questions:

(2 x 10)

- a) What is the disadvantage of dv/dt triggering of SCR?
- b) Differentiate between MOSFET & BJT?
- c) What is the power factor of a single phase full converter with highly inductive load for a firing angle of 45° ?
- d) What is secondary breakdown?
- e) What is the advantage of PWM rectifier over phase controlled rectifier?
- f) What are the difference between freewheeling diode and feedback diode?
- g) Differentiate between on-line and off-line UPS.
- h) Enumerate the difference between VSI & CSI.
- i) What are the difference between flyback converter and buck-boost converter?
- j) Explain the working of a Type-C chopper.

Q2 a) Explain the turn-off operation of GTO with relevant waveforms and describe the times associated with it. What are the advantages of GTO over SCR? (5)

b) Explain with relevant circuit diagram the V-I characteristics of UJT and the operation of a Synchronized UJT triggering circuit. (5)

- Q3 a)** A single phase fully controlled converter is operated from 230V, 50 Hz ac supply. The load current is continuous and ripple free with an average value of 5A. For firing angle of $\alpha = \pi/4$, determine (i) THD of input current (ii) Rectification Efficiency. **(5)**
- b)** Explain the various protection employed for thyristor with suitable circuit diagram and derive the expression for snubber circuit parameters. **(5)**
- Q4 a)** Draw and explain the operation of a three-phase Semiconverter with R-L-E load under continuous conduction with appropriate waveforms of (a) Output Voltage (b) Load current (c) phase-A supply current for a firing angle of 90° . **(5)**
- b)** A single phase full wave ac voltage controller has a R-L load of $R=10\Omega$ & $L=10\text{mH}$ and the input voltage is $V_s=230\text{V}$, 50 Hz. Determine (a) critical firing angle & (b) for a delay angle of 45° and extinction angle of 190° , the rms value of the output voltage. **(5)**
- Q5** What is soft switching? Describe the different modes of operation of a L-type ZCS Resonant Buck converter with proper circuit diagram and associated waveforms of inductor current and capacitor voltage. **(10)**
- Q6 a)** Explain the operation of Boost Regulator with suitable waveforms and derive the expression of average output voltage. **(5)**
- b)** A Boost Regulator has an input voltage of 10V dc. The average output voltage is 20 V and the average output current is 1 A. The switching frequency is 25 kHz. If the input side inductance $L=100\text{ }\mu\text{H}$ and output side capacitance $C=200\text{ }\mu\text{F}$, determine (i) The duty cycle (ii) The ripple current of inductor (iii) The ripple voltage of filter capacitor **(5)**
- Q7 a)** Draw and explain the 120° conduction mode operation of a 3- ϕ VSI with suitable waveforms of gate pulses, phase voltages & line voltages. **(5)**
- b)** Derive the expression of the RMS value of phase voltage, line voltage and the THD content of phase voltage for the above problem. **(5)**
- Q8** Write short notes on any two: **(5 x 2)**
- a)** Forward Converter
b) Battery Charging System
c) Electronic Ballast

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech
PEE5I101

5th Semester Regular / Back Examination 2018-19

POWER ELECTRONICS

BRANCH : ELECTRICAL

Time : 3 Hours

Max Marks : 100

Q.CODE : E300

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Short Answer Type Questions (Answer All-10) (2 x 10)

- Why circuit turn-off time is usually greater than thyristor turn off time ?
- What do you mean by latching current and holding current of a thyristor in operation.
- What is the electrical analogue of heat transfer from a power semiconductor device ?
- Can a power device be protected by a fuse? Justify.
- Single-phase converters are avoided in high performance applications. Why?
- What do you mean by AC link chopper? Draw the block diagram.
- The input voltage of a chopper is 220V, load voltage is 150V and the chopping frequency is 4Hz. Find the ON and OFF period of the thyristor in each cycle.
- What is the purpose of a converter in dc drives?
- How is the delay angle of one converter related to the delay angle of the other converter in a dual-converter system?
- What are the methods to vary the inverter output voltage? Write down some applications of inverter?

Part- II

Q2 Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Discuss the need of Snubber circuit and a series inductor for thyristor? Explain with clear circuit diagram.
- Explain the cosine waveform is used for designing a firing circuit?
- Enumerate the control circuit design for a two-quadrant chopper circuit.
- Derive the average output DC voltage for a single-phase controlled-bridge converter with both continuous and discontinuous current mode operations with relevant waveform.
- Derive the expression for the average output voltage for a three phase, 3-pulse and three phase, 6-pulse diode rectifier using cosine form.
- Ripple current is minimized by either increasing the chopper frequency or including an inductance in series with the armature of the dc motor. Discuss the merits and demerits of each alternative.
- What are the advantages of pulse width modulation? Describe the technique of technique of single pulse-width modulation with the expression for modulation index.
- How is SCR different from GTO? What are the merits and demerits of GTO?
- Draw and explain the current-voltage characteristic of a GTO thyristor switch and how it differs from an ordinary three wire thyristor switch ?
- Draw the circuit and explain the modified single phase series inverter with advantages and disadvantages.
- Draw the waveform of source voltage, source current, load voltage and load current of single phase full controlled converter for discontinuous load current with R-L load.

- l) What is cycloconverter? Explain the working of a single phase centre-tap cycloconverter with a simple diagram. Draw the input wave of frequency f and output wave of frequency $f/4$ for a resistive load.

Part-III

Long Answer Type Questions (Answer Any Two out of Four)

- Q3** a) Draw and explain the dynamic characteristic of thyristor during turn on and turn-off process. Draw the relevant circuit for this characteristic. **(10)**
- b) Find the protective elements of snubber circuit for protection of $\frac{di}{dt}$ and $\frac{dv}{dt}$ **(6)**
- Q4** a) Explain the operation with associated waveforms of a three-phase half wave controlled rectifier for a highly inductive load. Derive the expression for average and rms output voltage. **(10)**
- b) A three phase half wave controlled converter is operated from a three-phase Y-connected 208V, 60Hz supply and the load resistance is $R = 10 \text{ ohm}$. If it is required to obtain an average output voltage of 50% of the maximum possible output voltage, Calculate : (i) the delay angle α (ii) the rectification efficiency. **(6)**
- Q5** a) Draw and explain for three phase voltage source bridge type of inverter operating under 180° mode. **(10)**
- b) A first-quadrant dc-to-dc chopper feeds an inductive load of 10Ω resistance, 50mH inductance, and back emf of 55V dc, from a 340V dc source. If the chopper is operated at 200Hz with a 25% on-state duty cycle, determine, with and without (rotor standstill, $E = 0$) the back emf : **(6)**
- i) the load average and rms voltages;
- ii) the rms ripple voltage, hence ripple factor;
- iii) the maximum and minimum output current, hence the peak-to-peak output ripple in the current;
- iv) the average load output current.
- Q6** a) Draw the circuit diagram at different mode of operation for the single phase parallel inverter and explain the working with output voltage waveform. **(10)**
- b) Discuss the working function of a SMPS with block diagram. **(6)**

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech
PCEL4301

5th Semester Back Examination 2018-19
POWER ELECTRONICS
BRANCH : AEIE, EEE, EIE, ELECTRICAL, IEE
Time : 3 Hours
Max Marks : 70
Q.CODE : E379

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions: (2 x 10)

- a) What do you mean by latching current and holding current of an SCR?
- b) Differentiate between IGBT and MOSFET.
- c) Discuss the conditions that must be satisfied for turning on an SCR with gate signal.
- d) What do you mean by finger voltage?
- e) Which among string efficiency and derating factor should be more and why?
- f) What is zero crossing detector and why is it utilized in a triggering circuit?
- g) What is semi-converter? In what respect is the operation of single phase semi-converter different from that of full converter?
- h) A chopper is connected from a source terminal of 230V. The load voltage is of rectangular pulse duration of 1ms, over total time period of 5ms. Calculate the average and rms value of the load voltage.
- i) What is non-triggering voltage of an SCR?
- j) What is the significance of PWM technique in an inverter?

Q2 a) What is the need of Snubber circuit and a series inductor for a thyristor? Explain with clear circuit diagram (5)

b) Discuss on switching characteristics of GTO during turn-on and turn-off processes. (5)

Q3 a) A 50A thyristor is to be used in parallel with a 150A thyristor. The on-state voltage drops of the thyristors are 1.5V and 1.2V respectively. Calculate the series resistance R that should be connected with each thyristor if the two thyristors have to share the total current 200A into their rating. (5)

b) Discuss inverting mode of operation of single phase full wave controlled rectifier. (5)

Q4 a) A single phase ac voltage controller has input voltage 230V, 50Hz and a load of 10 ohms. Find the rms output voltage and draw relevant waveforms at a firing angle of 30°. (5)

b) What is cycloconverter? Explain the working of a single phase centre-tap cycloconverter with a simple diagram. Draw the input wave of frequency f and output wave of frequency f/4 for a resistive load. (5)

Q5 a) An R-L load, energized from single phase, 230V, 50Hz source through a single thyristor, has R=20 ohm and L=0.08 henry. If it is triggered in every positive half cycle at $\alpha=75^\circ$, find current expression as function of time. (5)

b) Derive the expression for average output voltage of a 3-phase full converter with a resistive load as a function of firing angle α . (5)

- Q6

a)

What is chopper? Explain different control strategies of chopper with relevant diagram.

(5)
- b)

With all the waveforms, explain the circuit operation of type-C chopper.

(5)

109

109

109

109

109

109

109

109

Q7

Draw and explain for three phase voltage source bridge type of inverter operating under 120° mode.

(10)

- Q8

Write short answer on any TWO :

(5 x 2)
- a)

UPS
- b)

Push Pull Converters
- c)

SPWM for voltage control
- d)

SVC

Registration No.:

--	--	--	--	--	--	--	--	--	--

Total Number of Pages: 02

Integrated Dual Degree (B.Tech and M.Tech)
REL4C003/REC4D002/REE4C003/REI4D002

4th Semester Regular/Back Examination: 2023-24

SUBJECT: Power Electronics

BRANCH(S): ELECTRICAL, ECE, ETC, EE, ECE, EEE, AEIE, EIE

Time: 3 Hour

Max Marks: 100

Q.Code: P410

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions: (2 x 10)

- Write the advantages of freewheeling diode.
- Write the drawbacks of resistant triggering method.
- Explain blanking time.
- What is the rms value of fundamental component of single phase bridge inverter if the input voltage is V .
- Explain why an anti-parallel diode is connected across each MOSFET in inverter.
- How is light triggering of a thyristor different from gate triggering?
- Why freewheeling diode is not required in a semi converter?
- What are the advantages of bipolar switching over unipolar switching in SPWM control strategy as applied to inverters?
- What is secondary breakdown of BJT?
- What is the importance of snubber circuit?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- A single-phase full wave rectifier with R-L-E load has $L = 6.5\text{mH}$, $R = 2.5\Omega$, and $E = 10\text{V}$. The input voltage is $V_s = 120\text{V}$ at 60Hz . Determine (I) the steady state load current at $\omega t = 0$, (II) the average diode current (III) the rms diode current (IV) the rms output current.
- The parameters of UJT are $V_s = 30\text{V}$, $\eta = 0.51$, $I_p = 10\mu\text{A}$, $V_v = 3.5\text{V}$, and $I_v = 10\text{mA}$. The frequency of oscillation is $f = 60\text{Hz}$, and the width of triggering pulse is $t_g = 50\mu\text{s}$. Assume $V_D = 0.5$. Design the triggering circuit.
- Explain with neat sketch the protection circuit of SCR.
- Describe the principle of step up chopper. Derive an expression for the average output voltage in terms of input voltage and duty cycle.
- Draw and explain the structure and characteristics power MOSFET and explain how it is different from BJT.

- f) A single-phase Semi converter feed power to RLE load. For discontinuous load current, draw the output voltage, load current, source current and freewheeling diode current waveforms as a function of time when
(I) Extinction angle $\beta > \pi$ (II) $\beta < \pi$ with $V_m \sin \beta < E$
- g) Draw and explain the dynamic characteristics of SCR during ON and OFF.
- h) For a step-down chopper, input dc voltage is 230V, load resistance=10 Ω . For the duty cycle of 0.5, Calculate I) the average and the rms values of output voltage and (II) Chopper frequency.
- i) Explain IGBT with its circuit symbol and switching characteristics.
- j) The input voltage to an SCR connected with snubber circuit is $V_s=200V$. The load and stray inductances are negligible and the thyristor is operated at a frequency of $f_s=2KHz$. If the required $(dv/dt) = 100 V/\mu s$, and the discharge current is to be limited to 100A, determine the value of RC Snubber circuit.
- k) A number of SCRs, each with rating of 2000V and 50A, are to be used in series-parallel combination in a circuit to handle 11KV and 400A. For a derating factor of 0.15, calculate the number of SCRs in series and parallel units. The maximum difference in their reverse recovery charge is 20 microcoulombs. Calculate (I) the value of dynamic equalizing capacitance and (II) the voltage across each of the slow thyristor in case one series connected SCR is fast.
- l) For a buck boost converter, the input dc voltage is 14V. The duty cycle is 0.6 with switching frequency of 25KHz. The inductance $L = 180\mu H$ and filter capacitance $C = 220 \mu F$. If the average load current is 1.5A, Compute
(i) The average output voltage
(ii) Peak to peak output voltage ripple
(iii) Peak to peak current in the inductor
(iv) The peak current of the device

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3 Explain 180° conduction scheme of a three-phase voltage source inverter with relevant circuit diagram and waveform. (16)
- Q4 With neat sketch explain the operation of three phase full wave ac to dc converter for R load with firing angle 15°. (16)
- Q5 Write short notes on (8x2)
(I) Sinusoidal PWM
(II) Buck Converter
- Q6 Explain the operations of four quadrant chopper. (16)

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

Course: B.Tech

Sub_Code: PEL5I101 / PEE5I101 / PET5J003

5th Semester Back Examination: 2022-23

SUBJECT Power Electronics

BRANCH(S): EEE, ELECTRICAL, ECE

Time : 3 Hour

Max Marks : 100

Q.Code : L573

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions : (2 x 10)

- What are the advantages of soft switched converters over hard switched converters?
- What are the advantages and disadvantages of MOSFETs ?
- What is harmonic factor and form factor in single phase half wave rectifier?
- What are the advantages of bipolar switching over unipolar switching in SPWM control strategy as applied to inverters?
- What are the merits and demerits of transformer gate isolation?
- What are the effects of load inductance on the performance of cycloconverter?
- What is the modulation index for a PWM control?
- Name three types of power supplies in general?
- What is the purpose of over modulation?
- What is a static VAR compensator?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- A single-phase bridge rectifier has a purely resistive load $R=10\Omega$, the peak supply voltage $V_m=170V$, and the supply frequency $f=60Hz$. Determine the average output voltage of the rectifier if the source inductance is negligible.
- Describe the basic principle of working of single phase to single phase step-down cycloconverter for continuous conduction for a bridge type cycloconverter.
- Discuss pulse-width modulation control of a converter.
- Explain the Cosine Firing Scheme for SCR.
- Describe the principle of step up chopper. Derive an expression for the average output voltage in terms of input voltage and duty cycle.
- Draw the Output and transfer characteristics of IGBTs.
- Describe flyback SMPS with relevant circuits and waveforms.

- h) Describe TRIAC firing circuit with relevant diagrams.
- i) Give Comparison between power MOSFET and BJT.
- j) What is the common method for dv/dt protection in thyristor?
- k) A single phase full bridge inverter is fed from a dc source such that fundamental component of output voltage is 230V. Find the rms value of thyristor and diode currents for resistive load of 2Ω .
- l) Draw the circuit diagram and output voltage waveform of zero voltage switching resonant inverter .

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** A 3-phase full converter thyristor bridge feeds a resistive load R. (16)
- (a) Sketch the input voltage waveforms
 - (b) Sketch the waveform of output current for a firing angle of 30°
 - (c) In case input voltage is 400V and $R=200\Omega$, indicate the peak magnitude of input current.
- Q4** Describe Cuk regulators with relevant equivalent circuits and waveforms. (16)
- Q5** A single phase bridge inverter has a resistive load of $R=2.4\Omega$ and dc input voltage is 48V. Determine (a) the rms output voltage at the fundamental frequency (b) The output power (c) the average and peak currents of each transistor (d) the peak reverse blocking voltage of each transistor (e) THD (f) The DF (g) The HF and DF of the LOH (16)
- Q6** Explain Current-Source Inverters in detail. (16)

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech

REC4D002 / REE4C003 / REI4D002 / REL4C003

4th Semester Regular / Back Examination: 2021-22

POWER ELECTRONICS

BRANCH(S): ECE, ETC / EEE, / AEIE / ELECTRICAL

Time : 3 Hour

Max Marks : 100

Q.Code : J614

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions :

(2 × 10)

- What is an inverter? List a few industrial applications of inverters.
- What is the need of PWM switching scheme in single phase inverter.
- What is the need of driver circuit in power switches?
- How does light triggering of a thyristor differ from gate triggering?
- What are the advantages of bipolar switching over unipolar switching in SPWM control strategy as applied to inverters?
- What is barrier potential? How are depletion layer and barrier potential effected by temperature.
- What are the advantages of bipolar switching over unipolar switching in SPWM control strategy as applied to inverters?
- Give five applications of phase controlled rectifier.
- Can you achieve zero degree firing angle for a SCR? Justify.
- State the difference between buck and boost converter.

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 × 8)

- What are the conditions for a thyristor to conduct?
- Define latching and holding currents as applicable to an SCR. Show these currents on its static I-V characteristics.
- Write short note on three-phase sinusoidal modulation.
- A step-up chopper has input voltage of 220V and output voltage of 660 V. If the conducting time of thyristor chopper is 100μs, compute the pulse width of output voltage. In case output voltage pulse width is halved for constant frequency operation, find the average value of new output voltage.
- What is a IGBT. Describe its basic structure.
- Distinguish between voltage commutation and current commutation in thyristor circuit.
- A single phase fully controlled rectifier has 220 sin (314t) as input supply voltage and resistor R as load. What will be the average output voltage for firing angle 30° for this rectifier?

- h) Describe the principle of DC-DC buck converter operation. Derive an expression for its average output voltage.
- i) Explain the concept of duty ratio in dc choppers. How it is used to control the output voltage.
- j) A single phase full converter delivers ripple free current to RL load with $R=15\Omega$. The source voltage is 230V, 50Hz. For firing angle of 30° , calculate rectification efficiency, displacement factor, power factor, THD, active and reactive powers.
- k) A single phase half wave converter is operated from a 120V, 60HZ supply and the load resistance is $R=10\Omega$. if the average output voltage is 25% of the maximum possible average output voltage, calculate (a) the delay angle (b) the rms and average output currents (c) the average and rms thyristor current and (d) the input power factor .
- l) Draw the load current waveform of single phase voltage source inverter with R-L load.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** A 3-phase full converter thyristor bridge feeds a resistive load R. (a) Sketch input voltage waveforms for V_{ab} , V_{ac} , V_{bc} etc. (b) Sketch the waveform of output and input current for phase A for firing angle of 30° . In case input voltage is 400V and $R=200\Omega$, indicate the peak magnitude of input current. **(16)**
- Q4** Explain the operation of single phase phase controlled full converter with highly inductive load with output waveforms. **(16)**
- Q5** What is pulse width modulation? List the various PWM techniques. How do these differ from each other? **(16)**
- Q6** Draw and explain the I-V characteristics of diode, MOSFET, and IGBT. **(16)**

--	--	--	--	--	--	--	--	--	--

5th Semester Regular / Back Examination: 2021-22**POWER ELECTRONICS**
Branch: ELECTRICAL/ EEE**Max Marks: 100****Time: 3 Hours****Q Code: OF231****Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.****The figures in the right hand margin indicate marks.****Part- I****Q1 Only Short Answer Type Questions (Answer All-10) (02×10)**

- What are the conditions for a thyristor to conduct? (2)
- What is the importance of snubber circuit? (2)
- What is freewheeling diode and what is its purpose? (2)
- What is the importance of blanking time in an inverter? (2)
- How does light triggering of a thyristor differ from gate triggering? (2)
- Can you achieve zero degree firing angle for a SCR? Justify. (2)
- Define latching current and holding current of a thyristor. (2)
- What are the advantages of bipolar switching over unipolar switching in SPWM control strategy as applied to inverters? (2)
- What are the advantages of soft switched converters over hard switched converters? (2)
- What are the benefits of cuk converters over buck boost converter? (2)

Part- II**Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (06×08)**

- Draw and explain the switching behavior of power MOSFET. (6)
- How is SCR protected against dv/dt and di/dt ? Explain with relevant circuit diagram. (6)
- A single phase 220V, 1KW electric room heater is connected across 220V, 50HZ supply through a triac. For a firing delay angle of 90° , calculate the power dissipated by the heater element. (6)
- For a buck boost converter the input dc voltage is 14V. The duty cycle is 0.6 with switching frequency of 25KHZ. The inductance $L=180\mu H$ and filter capacitance $C=220\mu F$. If the average load current is 1.5A, Compute
 - The average output voltage
 - Peak to peak output voltage ripple
 - Peak to peak current in the inductor
 - The peak current of the device
- Describe RC full wave triggering circuit for a single SCR. (6)
- Derive expressions for the average and r.m.s values of the output voltage waveform of single phase full wave converter. (6)
- Describe the principle of step up chopper. Derive an expression for the average output voltage in terms of input voltage and duty cycle. (6)
- A single phase full bridge rectifier having a supply voltage of $V_m \sin(\omega t)$ has a purely resistive load R. Determine
 - efficiency
 - the form factor
 - the ripple factor
 - the transformer utilization factor
 - the peak inverse voltage of diode
 - the crest factor of input current
- A single phase fully controlled rectifier has $200 \sin(314t)$ as input supply voltage and resistor R as load. What will be the average output voltage for firing angle 30° for this rectifier? (6)
- The parameters of UJT are $V_s = 30V$, $\eta = 0.51$, $I_p = 10\mu A$, $V_v = 3.5V$ and $I_v = 10mA$. The frequency of

oscillation is $f=60\text{HZ}$, and the width of triggering pulse is $t_g = 50\mu\text{s}$. Assume $V_D =0.5$. Design the triggering circuit .

k) Explain the operation of SMPS. (6)

l) What is a GTO. Describe its basic structure. (6)

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four) (02×16)

A single-phase Semi converter feed power to RLE load. For discontinuous load current, draw the output voltage, load current, source current and freewheeling diode current waveforms as a function of time when (16)

(i) Extinction angle $\beta > \Pi$ (ii) $\beta < \Pi$ with $V_m \sin \beta < E$

Write short notes on (16)

(i) Sinusoidal PWM (16)

(ii) Comparison between power MOSFET and BJT

Explain 180° conduction scheme of a three phase voltage source inverter with relevant circuit diagram and waveform. (16)

Describe the circuit operation of four-quadrant chopper with relevant circuit diagrams and its operation in all the four quadrants . (16)

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech

REC4D002/REE4C003/REI4D002/REL4C003

4th Semester Regular / Back Examination: 2022-23

SUBJECT: Power Electronics

BRANCH(S): ECE,ETC, EEE, AEIE,EIE, ELECTRICAL

Time : 3 Hour

Max Marks : 100

Q.Code: M277

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions:

(2 x 10)

- State different types of diode and their applications.
- What are the advantages of bipolar switching over unipolar switching in SPWM control strategy as applied to inverters?
- What is the need of driver circuit in power switches?
- In a dc chopper, the average load current is 30 Amps, chopping frequency is 250 Hz. Supply voltage is 110 volts. Calculate the ON and OFF periods of the chopper if the load resistance is 2ohms.
- What is a secondary breakdown of BJT?
- Define latching current and holding current of a thyristor.
- What is the importance of blanking time in an inverter?
- What is the purpose of over modulation?
- What is an IGBT? Describe its basic structure.
- What are the merits and demerits of transformer gate isolation?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Draw and explain the switching behavior of power BJT.
- Explain gate Protection of Thyristor.
- The parameters of UJT are $V_s = 30\text{ v}$, $\eta = 0.51$, $I_P = 10\mu\text{A}$, $V_V = 3.5\text{V}$ and $I_V = 10\text{mA}$. The frequency of oscillation is $f = 60\text{Hz}$, and the width of triggering pulse is $t_g = 50\mu\text{s}$. Assume $V_D = 0.5$. Design the triggering circuit.
- Draw and explain the structure and characteristics power MOSFET and explain how it is different from BJT.
- Describe the principle of DC-DC buck converter operation. Derive an expression for its average output voltage.

- f) A three phase half wave converter is supplying a load with a continuous constant current of 50A over a firing angle from 0° to 60° . What will be the power dissipated by the load at these limiting values of firing angle. The supply voltage is 415V (line).
- g) How is SCR protected against dv/dt and di/dt ? Explain with relevant circuit diagram.
- h) A single phase fully controlled rectifier has $250 \sin(314t)$ as input supply voltage and resistor R as load. What will be the average output voltage for firing angle 40° for this rectifier?
- i) Draw the V-I and switching characteristics of thyristor.
- j) A single PWM full bridge inverter feeds an RL load with $R = 10\Omega$ and $L = 10 \text{ mH}$. If the source voltage is 120V, find out the total harmonic distortion in the output voltage and in the load current. The width of each pulse is 120° and the output frequency is 50Hz.
- k) A step up chopper has input voltage of 220V and output voltage of 660V. If the conducting time of thyristor-chopper is $100\mu\text{s}$, Compute the pulse width of output voltage. In case output voltage pulse width is halved for constant frequency operation, find the average value of new output voltage.
- l) Describe the operation of single phase voltage source inverter.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** A single phase bridge inverter has a resistive load of $R = 2.5\Omega$ and dc input voltage is 50V. Determine (a) the rms output voltage at the fundamental frequency (b) The output power (c) the average and peak currents of each transistor (d) the peak reverse blocking voltage of each transistor (e) THD (f) The DF (g) The HF and DF of the LOH. (16)
- Q4** Explain in detailed the unipolar and bipolar switching schemes. (16)
- Q5** Write short notes on any two (16)
 - (i) Sinusoidal PWM
 - (ii) Comparison between power MOSFET and BJT
 - (iii) ZCS resonant Inverter
 - (iv) Three-phase sinusoidal modulation
- Q6** Explain the operation of 3 phase controlled full converter with R-L load with output waveforms. (16)

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 01

B.Tech
PCEL4301

5th Semester Back Examination 2019-20

POWER ELECTRONICS

BRANCH : AEIE, EEE, EIE, ELECTRICAL, IEE

Time : 3 Hours

Max Marks : 70

Q.CODE : HB284

Answer Question No.1 which is compulsory and any FIVE from the rest.

The figures in the right hand margin indicate marks.

Q1 Answer the following questions : (2 x 10)

- Draw the V-I characteristics of IGBT.
- What are the advantages of using UJT triggering scheme?
- How dv/dt protection is achieved in SCR?
- Define input current harmonic factor.
- What are the advantages of 120° conduction mode over 180° conduction mode?
- What is space vector modulation?
- What do you mean by hard-fired thyristor?
- Explain constant frequency system control of dc-dc converter.
- What is the main drawback of half-bridge inverter?
- What are the advantages of using PWM inverter?

Q2 a) A single-phase full bridge diode rectifier is supplied from 230V, 50 Hz source. The load consists of resistance of $R=10\Omega$ and a large inductance so as to render the load current constant. Determine : (i) average values of output voltage and output current, (ii) rms values of output and input currents, and supply p.f. . **(5)**

b) Draw the output waveform of three-phase diode bridge rectifier with R-L load. **(5)**

Q3 a) Draw and explain the static V-I characteristics of GTO. **(5)**

b) Explain the switching characteristics of GTO with neat sketch waveform. **(5)**

Q4 a) Explain the operation of single phase full wave controlled rectifier. **(5)**

b) A single-phase full converter bridge is connected to RLE load. The source voltage is 230V, 50 Hz. The average load current of 10A is constant over the working range. For $R=0.4\Omega$ and $L=2\text{mH}$, compute: (i) firing angle delay for $E=120\text{V}$, (ii) firing angle delay for $E=-120\text{V}$. Indicate which source is delivering power to Load in part (i) and (ii). **(5)**

Q5 a) Explain the principle of operation of three phase PWM rectifier. **(5)**

b) A type-A chopper has input dc voltage of 200V and a load of $R=10\Omega$ in series with $L=80\text{mH}$. If load current varies linearly between 12A and 16A, find the time ratio $T_{\text{on}}/T_{\text{off}}$ for this chopper. **(5)**

Q6 Describe in detail the 180° conduction mode of the voltage source inverter with neat sketch diagram and waveform. **(10)**

Q7 Discuss the operation of forward converter and push pull converter. **(10)**

Q8 Write short Notes on any TWO : (5 x 2)

- Current source inverter
- Zero voltage switching resonant inverter
- Static VAR Compensator

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech
PEE51101

5th Semester Regular / Back Examination 2019-20

POWER ELECTRONICS

BRANCH : ELECTRICAL

Max Marks : 100

Time : 3 Hours

Q.CODE : HRB074

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) (2 x 10)

- Draw the V-I characteristics of power diode.
- Draw the V-I characteristics of thyristor.
- Draw the switching characteristics of SCR of anode current.
- Why series and parallel connection of SCR are required ?
- Why Cosine firing scheme is used ?
- Derive the average output voltage of single phase half wave controlled rectifier with R-L load.
- Derive the average output voltage of single phase Full wave controlled rectifier with R-L load.
- Why GTO is preferred over SCR ?
- What is Time Ratio Control of DC-DC converter ?
- Why 180° mode of conduction is preferred over 120° mode of conduction?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Discuss the steady state characteristics of BJT.
- Find the time required to deliver a charge of 200Ah through a single-phase half-wave diode rectifier with an output current of 100A rms and with sinusoidal input voltage. Assume diode conduction over half cycle.
- A single -phase full bridge diode rectifier is supplied from 230 V, 50 Hz source. The load resistance of $R = 10\Omega$ and a large inductance so as to render the load current constant. Determine: (i) average values of output voltage and output current, (ii) average and rms values of diode currents.
- A 3-phase bridge rectifier charges a 240V battery. Input voltage to rectifier is 3-phase, 230V, 50 Hz. Current limiting resistance in series with battery is 8Ω and an inductor makes the load current almost ripple free. Determine power delivered to the battery and load.
- Explain the static I-V characteristics of a thyristor.
- Explain and derive the design of a snubber circuit for a thyristor controlling the power in a load resistance R_L . The supply voltage is 240V dc and the specified limits for di/dt and dv/dt for the SCR are 50 A/ μ sec and 300 V/ μ sec respectively. Determine the values of the di/dt inductance and the snubber circuit parameters R_s and C_s .
- Explain the effect of source impedance on the performance of converters with neat sketch diagram and expressions.

- h) Explain the voltage control of 3-phase inverter by sinusoidal PWM scheme.
- i) An RLE load is operating in a chopper circuit from a 500-volt dc source. For the load, $L=0.06H$, $R=0$ and constant E . For a duty cycle of 0.2, find the chopping frequency to limit the amplitude of load current excursion to 10A.
- j) A single-phase full-wave ac voltage controller feeds a load of $R=20\ \Omega$ with an input voltage of 230V, 50 Hz. Firing angle of both the thyristor is 45° . Calculate (a) rms value of output voltage (b) load power and (C) input power factor.
- k) Explain the principle and operation of single-phase mid-point type step-up cycloconverter.
- l) Explain the operation of current source inverter with neat sketch diagram and waveforms.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** Draw and Explain in details three-phase 180° mode of VSI. **(16)**
- Q4** Explain the applications of UPS, SMPS, Battery Chargers and SVC. **(16)**
- Q5** Discuss the switching characteristics of an SCR during turn ON and turn OFF with neat sketch waveform. **(16)**
- Q6** Discuss the operation of Fly Back converter and Push Pull Converter with neat sketch diagram and waveforms. **(16)**

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech
PCEL4301

3rd Year Special Examination 2018-19
POWER ELECTRONICS

BRANCH : AEIE, EEE, EIE, ELECTRICAL

Time : 3 Hours

Max Marks : 70

Q.CODE : G918

Answer Question No.1 which is compulsory and any FIVE from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions :

(2 x 10)

- a) State at least two applications of SCR.
- b) Draw the transfer characteristics of N-channel enhancement type MOSFETs.
- c) What is a half controlled rectifier?
- d) Write the output voltage equation of three phase fully controlled rectifier.
- e) State at least two applications of chopper.
- f) Draw the circuit diagram of class-C chopper.
- g) State at least two applications of AC voltage regulator.
- h) What is a push pull converter?
- i) Draw a simple battery charging circuit.
- j) What are the advantages of sinusoidal PWM?

Q2 a) Describe the switching characteristics of power MOSFETs and IGBT.

(5)

b) Explain the need of commutation in thyristor circuits. What are the different methods of commutation scheme?

(5)

Q3 a) Explain the operation of single phase fully controlled converter. Derive the output voltage and current waveforms.

(5)

b) A 220V DC motor has a speed of 1000 rpm at no load. This motor is fed through a single phase fully controlled converter. Find the firing angle, when motor runs at 724 rpm and draws a full load current of 20 A. Assume ripple free armature current. Armature resistance is 1 ohm and supply voltage is 230V AC.

(5)

Q4 a) What is meant by step-up chopper? Derive the output voltage equation of Step up chopper and draw the necessary waveforms

(5)

b) A chopper is feeding an RL load with $V_s=220$ V, $R=5$ ohms, $L=7.5$ mH and the chopper frequency is 1 kHz. Duty cycle is 0.5. Calculate :

(5)

- (i) the minimum and maximum instantaneous load current
- (ii) peak to peak ripple load current

Q5 a) Describe the working of a three-phase voltage source inverter with 180° conduction scheme. Draw the appropriate circuit diagram and waveforms?

(5)

b) What is meant by step-up and step-down cycloconverter? Explain its operation. Sketch the input voltage, output voltage waveforms in reference to the drive to the switch.

(5)

Q6

Examine the structure and different modes of operation with the characteristics of TRIAC.

(10)

Q7

A star connected three phase balanced R-L load with $R=10\text{ Ohms}$ and $L=30\text{ mH}$ is connected to a three phase inverter. Input voltage is 300 V , frequency= 50 Hz with 180° conduction scheme. Find :

(10)

a) Output voltage,

b) Power delivered to the load,

e) distortion factor.

b) Output current,

d) harmonic factor

Q8

Write short answer on any TWO :

(5 x 2)

a) Flyback converter

b) Current Source Inverter

c) SMPS

Registration No :

--	--	--	--	--	--	--	--	--	--

Total Number of Pages : 02

B.Tech
PEE5I101

3rd Year Special Examination 2018-19

POWER ELECTRONICS

BRANCH : ELECTRICAL

Max Marks : 100

Time : 3 Hours

Q.CODE : G815

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10)

(2 x 10)

- What are the different methods to turn on the thyristor?
- How can a thyristor turned off?
- What are the advantages of GTO over SCR?
- What is meant by delay angle?
- Write down the expression for average output voltage for step up chopper.
- What are the advantages of single phase bridge converter over single phase dpont converter?
- What are the disadvantages of the harmonics present in the inverter system?
- What are the disadvantages of AC voltage controllers?
- What are the applications of Cycloconverter?
- What type of gating signal is used in single phase AV voltage controller with RL load?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)

(6 x 8)

- Compare any five salient features of MOSFET with IGBT.
- Explain the operation of single phase dual converter with circulating and non-circulating current type.
- Explain the operation of three phase semi converter with RL load. Sketch the associated waveforms.
- Draw the circuit diagram of buck regulator and explain its working principle with necessary waveforms.
- If the firing angle of a single phase half-wave thyristor controlled rectifier with purely resistive load is $\pi/2$ radians. Determine :
 - the rectification efficiency
 - the transformer utilization factor (UTF)
 - the peak inverse voltage (PIV) of the thyristor.
- For a SCR, gate-cathode characteristic is given by $V_g = 1+10 I_g$. Gate source voltage is a rectangular pulse of 15V with 20 μ S duration. For an average gate power dissipation of 0.3 W and peak gate drive power of 5W, compute
 - The series resistance to be connected in series with the SCR gate
 - The triggering frequency and
 - The duty cycle of the triggering pulse
- Explain sinusoidal and multiple PWM techniques used in inverter.
- Explain the principle of working of single phase to single phase step up cycloconverter.
- Describe the working principle of buck-boost converter with necessary circuit and waveforms.
- Explain the various control strategies of chopper.
- Write a short note on Push Pull SMPS.
- Explain zero voltage switching resonant converters.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 With necessary circuit and waveforms, explain the principle of operation of three phase controlled bridge rectifier feeding R-L load and derive the expression for the average output DC voltage. (16)

Q4 A 3 phase bridge full converter is used to control a dc motor. Input voltage is 200V (line to line), motor's back emf is 220V and armature resistance is 1 Ohm. Find the following quantities if continuous and ripple free load current is 25A : a) Output voltage b) Firing angle (16)

Q5 Explain the principle of operation of 3 phase voltage source inverter with 180°conduction mode with necessary waveforms and circuits. Also obtain the expression for line to line voltage. (16)

Q6 An AC voltage controller has a resistive load of $R=10$ ohms and the R.M.S. input voltage is $V_s=120$ V, 60 Hz. The thyristors switch is on for $n=25$ cycles and is off for $m=75$ cycles. Determine : (16)
a) The R.M.S. output voltage V_o ,
b) The input power factor.